INDUSTRIAL TRAINING REPORT

BIT40112

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DECLARATION

I sincerely declare that

1. I am the sole writer of this report.
2. The details of training and experience contain in this report describe my involvement as a trainee in the field of computer science/information technology.
3. The report is only prepared for my academic requirement not for any other purpose.
4. All the information contains in this report is certain and correct to the knowledge of the author.

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ABSTRACT

Industrial training is an important component of information technology education as it provides exposure to real-world working environments and practical application of theoretical knowledge. This report describes the industrial training experience conducted at Pemaju Digital Sdn. Bhd., a digital solutions provider based in Selangor, Malaysia, specializing in website development, system development, digital marketing, and domain services. During the training period, various daily tasks were performed, including the development and maintenance of WordPress landing pages, application of basic search engine optimization (SEO) techniques, DNS verification for company-managed domains, and assistance in developing automotive prospect management systems using a local development environment. These activities enhanced understanding of website management, digital marketing practices, and system workflows in a professional setting. In addition to daily operational tasks, a student-proposed project known as the DomTrack Management System was developed to address inefficiencies in the existing Excel-based approach used for managing domain and cPanel records. The manual method previously applied resulted in data duplication, outdated records, security risks, and inefficient searching processes. To overcome these limitations, DomTrack was developed as a web-based system using Laravel, MySQL, and Bootstrap, following the Waterfall development methodology. The system centralizes domain data, applies role-based access control, and provides features such as domain management, invoice generation, and PDF report export to improve operational efficiency and data accuracy. System testing confirmed that all modules functioned correctly, and the system was successfully deployed using cPanel for internal company use. In conclusion, the industrial training experience strengthened technical competencies in web-based system development and enhanced professional skills such as time management and problem-solving.

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CHAPTER 1

INTRODUCTION

1.1 Organization Background

Pemaju Digital Sdn. Bhd., established in 2009 and located in Selangor, Malaysia, is a digital solutions provider specializing in website design and development, online systems, e-commerce solutions, hosting and domain services, and digital marketing, including search engine optimization (SEO) and social media advertising. The company also offers creative services such as graphic design, photography, and videography to support client projects. These services enable clients, particularly small and medium enterprises (SMEs), to develop a strong online presence and maintain competitive advantage in the digital marketplace.

Over the years, the company has expanded its operations to provide a one-stop solution for digital needs. Projects are handled from concept design to deployment and marketing, ensuring a seamless delivery of services. The team at Pemaju Digital consists of experienced professionals in web development, design, marketing, and project management, allowing the company to meet client expectations and adapt to emerging technologies.

The company's vision is to become a leading digital solutions agency in Malaysia, recognized for innovative, user-centric, and high-performance services. Its mission emphasizes providing tailored digital solutions that address client needs, supporting business growth while maintaining a commitment to quality, efficiency, and professionalism. Continuous service expansion and adaptation reflect the company's commitment to evolving market trends and fostering long-term client relationships.

1.2 Vision and Mission of the Company

The vision of Pemaju Digital Sdn. Bhd. is to become a reliable digital solution provider that delivers practical and effective systems to support business growth.

The mission of the company includes developing user-friendly systems, providing reliable technical support, and continuously improving service quality through innovation and technology. The company also aims to maintain long-term relationships with clients by offering scalable and maintainable digital solutions.

1.3 Organization Structure

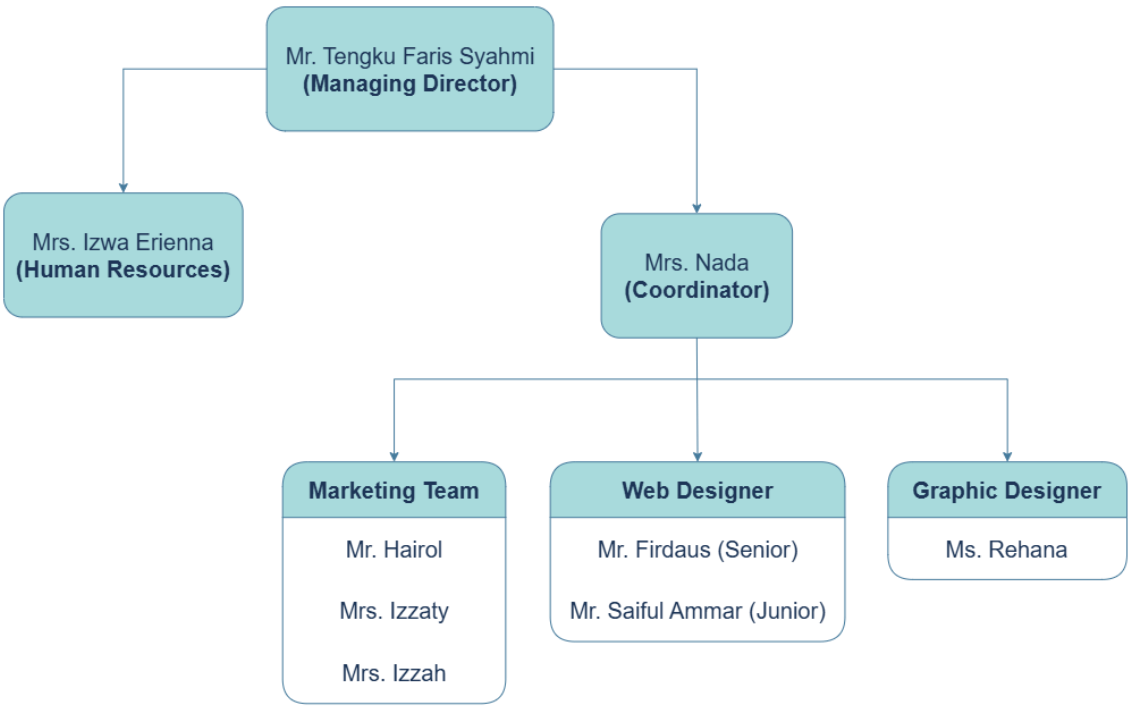


Figure 1.1: Pemaju Digital Sdn. Bhd. Organization Structure

Figure 1.1 shows the organizational structure of Pemaju Digital Sdn. Bhd. is built in a simple and clear hierarchy to ensure smooth operations. At the top of the structure is the Managing Director, who is responsible for the overall direction of the company. This role includes making major decisions, guiding the company's goals, and monitoring all departments.

Below the Managing Director is the Human Resources Officer, who handles staff matters, attendance, discipline, and administrative tasks. This position ensures that workplace policies are followed and that the team receives proper support in their daily work. The

coordinator plays an important role as the link between departments. This position manages daily operations and ensures that tasks are carried out on schedule. The coordinator supervises several teams, including the Marketing Team, Web Designers, and the Graphic Designer, to make sure all work is aligned and organized.

The Marketing Team consists of three staff members. They are responsible for promoting the company's services, managing clients, preparing proposals, and handling communication with customers. Their work helps maintain the company's presence and reputation in the digital market. The Web Design department includes a Senior Web Designer and a Junior Web Designer. The senior designer oversees technical development and ensures quality, while the junior designer assists with website layout, content updates, and system improvements. Together, they handle all website-building tasks for clients. The Graphic Designer manages visual materials such as banners, logos, posters, and social media graphics. This role supports both internal projects and client needs, ensuring strong and consistent visual branding. This organizational structure allows Pemaju Digital to maintain clear communication, efficient workflow, and effective coordination between all departments.

1.4 Product and Service Overview

Pemaju Digital Sdn. Bhd. provides a wide range of digital products and services designed to support businesses in establishing and growing their online presence. The company positions itself as a one-stop digital agency, offering integrated solutions that combine marketing, system development, and online business tools. With experience serving more than 11,000 customers across Malaysia, the services are structured to meet the needs of small, medium, and large-scale businesses from various industries.

One of the core services offered is website design and development. This includes corporate websites, e-commerce platforms, and customised systems tailored to client requirements. The websites are developed with a focus on clean layout, user-friendly navigation, and responsive design to ensure accessibility across different devices. In addition, Pemaju Digital also provides system development services, such as customer management systems, booking systems, and internal business systems, which help clients manage data and operations more efficiently.

In terms of digital marketing solutions, Pemaju Digital offers services such as online marketing strategies, branding support, and performance-focused digital campaigns. These services aim to help businesses increase visibility, attract potential customers, and improve overall online engagement. The company emphasizes the use of latest proven strategies that are suitable for the Malaysian market, ensuring that marketing efforts align with current trends and consumer behaviour.

Other supporting services include graphic design, premium domain sales, and technical support and maintenance. Graphic design services cover branding materials such as logos, banners, and promotional visuals, while premium domain services help businesses secure suitable domain names for branding purposes. Continuous support, client communication through multiple channels, and structured working hours further enhance service reliability and customer satisfaction. Overall, the products and services offered by Pemaju Digital reflect a comprehensive digital solution approach focused on quality, efficiency, and long-term client value.

CHAPTER 2

INDUSTRIAL TRAINING DAILY TASKS & PROJECT

2.1 Industrial Training Daily Tasks

Daily tasks during the industrial training at Pemaju Digital Sdn. Bhd. focused on website management, digital marketing practices, and prototype system development. Activities included creating and maintaining WordPress landing pages for clients, such as customizing themes, updating content, ensuring responsive layouts, and maintaining a visually appealing and user-friendly interface. These tasks contributed to improving practical skills in website management and understanding client requirements in real-world projects.

In addition, SEO strategies were applied to enhance website performance, including keyword research, content optimization, monitoring page analytics, and implementing on-page and off-page SEO techniques. These activities helped strengthen knowledge in digital marketing and improved the ability to support clients in achieving better online visibility and engagement.

Development of new prospect management systems for automotive brands such as Honda, Perodua, Proton, and Toyota was also carried out using Laragon as the local development environment. This included creating data structures, managing records, and understanding workflow requirements for automotive prospect management. Additionally, DNS verification for company-managed domains was performed to ensure proper configuration of A records and NS records, maintaining website functionality and email delivery.

The combination of website development, SEO optimization, prototype system development, and DNS verification contributed to building technical skills while supporting ongoing company operations. These daily tasks also provided practical preparation for the development of the DomTrack Management System, allowing knowledge gained from routine activities to be applied to larger, structured projects.

2.2 Student-Proposed Project: DomTrack Management System

DomTrack is a web-based management system developed during the industrial training period to improve the management of domain and cPanel records within the company. The system was proposed to replace the existing Excel-based method, which had become increasingly inefficient as the number of managed domains and hosting accounts continued to grow. Manual data handling through spreadsheets caused difficulties in tracking expiry dates, login credentials, and provider details, while also increasing the risk of duplicated records, outdated information, and human error.

Before the introduction of DomTrack, domain and cPanel information was stored across multiple Excel files. This approach made data searching and updating time-consuming, as staff needed to manually check several files to locate accurate information. The lack of a centralised system also made it difficult to ensure data consistency, especially when multiple users accessed and edited the same records. These limitations highlighted the need for a more structured and reliable system to support daily operational tasks.

DomTrack centralises all domain-related information into a structured MySQL database, allowing data to be stored in an organised and secure manner. Each domain record contains essential details such as domain name, cPanel access, provider credentials, expiry date, and system status. With all information stored in a single platform, data retrieval becomes faster and more accurate compared to manual spreadsheet management.

The system also implements role-based access control to ensure proper data security and system usage. Administrators are granted full access to manage domain records, user accounts, and system settings, while staff users are limited to viewing records and generating reports based on assigned permissions. This access structure helps prevent unauthorised data modification and ensures sensitive information is handled responsibly.

In addition, DomTrack includes search and filter functions, report generation features, and a clear user interface to support daily operations. These features allow easier monitoring of active and expired domains, reduce administrative workload, and improve data accuracy. The development of DomTrack involved a collaborative team with clearly defined roles, as shown in Table 2.1, and supports a more structured, consistent, and maintainable domain management process within the organisation.

Table 2.1: Team members involved

Team Member	Role	Responbilities
Nurfazrina binti Azmi	Developer A	• Managed domain listing, search functionality, and addition of new entries
Muhammad Norkhaiqal bin Norhaidi	Developer B	• Responsible for login and authentication modules, as well as the main dashboard
Nabil Syahmi bin Sahizan	Developer C	• Responsible for report and invoice generation, as well as PDF export,

2.3 Problem Statement

Prior to the development of DomTrack, domain and cPanel information was managed using multiple Excel spreadsheets. As the number of domains increased, this method became difficult to maintain and prone to errors. Data duplication, outdated records, and inconsistent formatting frequently occurred due to manual updates and the absence of a centralised system.

Searching for specific domain information required staff to manually check several files, which was time-consuming and inefficient. Tracking important details such as expiry dates and login credentials was also challenging, increasing the risk of missed renewals and incorrect data usage. These issues affected daily operations and reduced overall efficiency in managing client domains.

Furthermore, the Excel-based approach did not provide proper access control. Sensitive information such as provider credentials and cPanel logins could be accessed or modified without restriction, posing potential security risks. The lack of structured reporting and filtering functions also limited the ability to monitor domain status effectively.

These challenges highlighted the need for a dedicated web-based system that could centralise domain records, improve data accuracy, enhance security, and support efficient searching and reporting. DomTrack was therefore developed to address these problems by providing a structured, secure, and user-friendly platform for domain and cPanel management.

2.4 Project Objectives

The objectives of this system are to:

i. Design a structured domain management system using an organised and object-oriented approach to replace manual Excel handling.
ii. Develop a web-based application for managing domain and cPanel records using Laravel, MySQL, and Bootstrap to ensure better accessibility and workflow efficiency.
iii. Test the developed system to ensure all modules function correctly, including data entry, searching, updating, and role-based access control.

2.5 Project Scope

The scope of this project focuses on the development of DomTrack, a web-based system designed to manage domain and cPanel records used by the company. The system is developed to replace the previous Excel-based approach, which caused issues such as duplicated records, inconsistent data updates, and difficulty in tracking the latest domain information. The scope is limited to internal use and is intended to support daily operational tasks related to domain management, invoicing, and reporting.

Table 2.2: List of System Users

System User	Scope
Administrator	• Log in to the system • Manage domain and cPanel records • Generate and manage invoices • Generate domain reports in PDF format • Manage staff accounts and system settings • View dashboard summary
Staff	• Log in to the system • View dashboard information • View system settings (read-only) • Generate and download invoices

Table 2.2 shows the users involved in this system are limited to internal staff members of the company. Each user role is defined clearly to avoid unauthorized access and to ensure that sensitive information such as domain login credentials is handled responsibly. The scope does not include integration with external systems, public user access, or payment gateway functionality.

Table 2.3: List of System Functional Modules

Functional Module	Description
Authentication	• Allows users to log in and log out of the system securely • Controls access based on user roles (Admin and Staff)
Dashboard	• Displays summary information such as total domains, active domains, and expiring domains
Domain Management	• Displays summary information such as total domains, active domains, and expiring domains • Stores domain details including credentials, provider information, expiry dates, and cPanel links
Invoice Management	• Enables creation, editing, deletion, and downloading of invoices • Generates invoices in PDF format using stored domain data
Report Generation	• Produces domain summary reports • Allow reports to be exported in PDF format

Table 2.3 shows the system covers essential functions required for managing domain data in a structured and controlled environment. These include storing domain credentials, monitoring expiry dates, generating invoices, and producing domain reports in PDF format. Access to the system is controlled through user roles to ensure data security and proper task separation. Only authorized users are allowed to perform specific actions based on their assigned role.

2.6 Tools and Software

The development of the DomTrack Management System involved the use of several tools and software to support system design, development, testing, and deployment. These tools were selected based on their suitability for web-based application development and their common usage within the company's working environment. The combination of these technologies ensured that the system was stable, maintainable, and aligned with industry practices.

Laravel framework was used as the main backend development platform. Laravel provides a structured MVC architecture that helps organise system logic, database interaction, and user interface handling in a clear manner. Composer was used as a dependency management tool to install and manage Laravel packages required during development. This allowed external libraries to be integrated efficiently while maintaining version control and system compatibility.

For the frontend interface, Bootstrap was implemented to create a responsive and user-friendly layout. Bootstrap components such as forms, tables, buttons, and navigation bars were used to ensure consistent design across different system pages. Visual Studio Code (VS Code) was used as the primary code editor due to its lightweight performance, syntax highlighting, and support for Laravel and PHP development.

The system database was managed using MySQL through phpMyAdmin, which allowed easier database creation, table management, and data inspection during development and testing. Git was used for version control to track changes and manage collaborative development. Finally, cPanel was utilised for deployment and hosting management, enabling the system to be accessed in a live environment and supporting real-world usage within the company's operations.

2.7 Project Methodology

The development of DomTrack followed the Waterfall methodology. This approach was chosen because the system requirements were clearly defined at an early stage, allowing development to progress in structured phases. Figure 2.1 shows Waterfall Model structure.

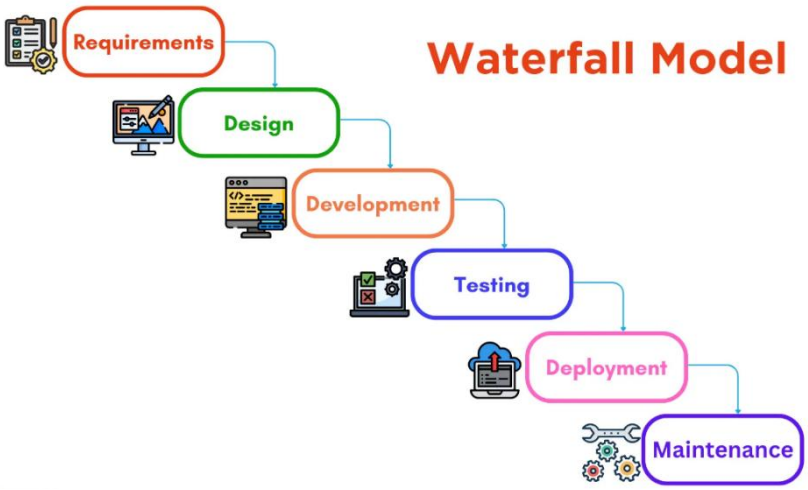


Figure 2.1: Waterfall Model

The Waterfall approach allowed each stage to be completed before moving to the next. Requirements were analyzed first to understand the company's needs. Design involved database planning, page layouts, and module structures. Development was done using Laravel, MySQL, and Bootstrap, followed by testing to fix errors and improve functionality. Deployment involves moving the system to the production server, and Maintenance ensures the system remains stable and updated after deployment.

This methodology was suitable for DomTrack because the project scope was clear, and tasks could be completed sequentially. Using Waterfall helped maintain a structured workflow, ensured proper documentation, and made it easier to track progress and resolve issues at each stage.

CHAPTER 3

SYSTEM DEVELOPMENT (WATERFALL MODEL)

3.1 Requirement Phase

The first stage of the Waterfall methodology involved gathering all necessary requirements for DomTrack. Initially, the current workflow of managing domain and cPanel records using Excel sheets was studied in detail. This included understanding how records were stored, the types of data fields required, and common problems such as duplicate entries and outdated information. Interviews and discussions with supervisors and staff helped identify what features were essential, including user roles, search functions, and report generation.

At this stage, all functional and non-functional requirements were documented. Functional requirements focused on modules like Authentication, Domain Management, Reporting, and Invoice Generation, while non-functional requirements addressed system performance, security, and accessibility. To visualize the system scope, a Context Diagram was created showing the interactions between users (Admin and Staff) and the system. This Diagram highlights the boundaries of the system and its primary external entities, and it can be attached as Figure 3.1 in the report.

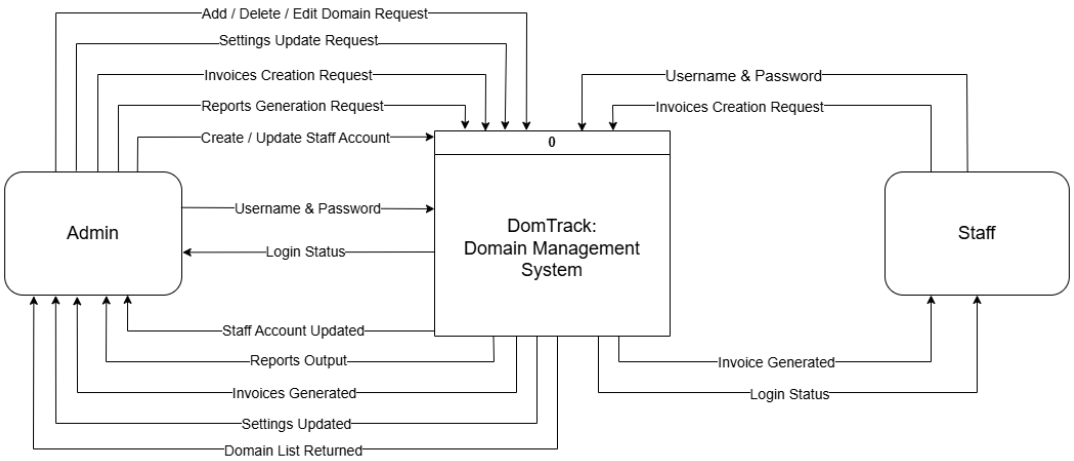


Figure 3.1: Context Diagram

The context diagram illustrates the interaction between the system, the Admin, and the Staff. The Admin performs several key functions, including managing domain information, updating system settings, creating invoices, generating reports, and creating or updating staff accounts. For each action, the system sends back the relevant output, such as updated staff records, generated invoices, completed reports, updated settings, and the domain list. The Admin also submits login credentials, and the system returns a login result, either granting access or rejecting the attempt.

For the Staff, the interaction is more limited. They submit their login details and can request invoice creation. In response, the system returns the generated invoice and the login result. This diagram highlights the main inputs and outputs exchanged between each user group and the system at a high level.

3.2 Design Phase

Before starting the detailed design, the team focused on planning how the system would function and how data would be organised. This included designing the database, mapping system processes, and preparing layouts for key pages. The aim was to ensure that every module, data flow, and user interaction was clearly defined, so development could proceed smoothly and efficiently. Diagrams such as DFDs, ERD, flowcharts, and wireframes were prepared to support this planning and provide a visual guide for the team.

3.2.1 Database Design

The database design for DomTrack was developed to support all core system functions, including domain management, user authentication, and invoice generation. A relational database model was chosen to ensure data integrity, minimize redundancy, and allow efficient querying, especially for reporting and dashboard features. This section explains the entities, attributes, and relationships illustrated in the Entity Relationship Diagram (ERD). Figure 2.3 shows the ERD for the DomTrack system.

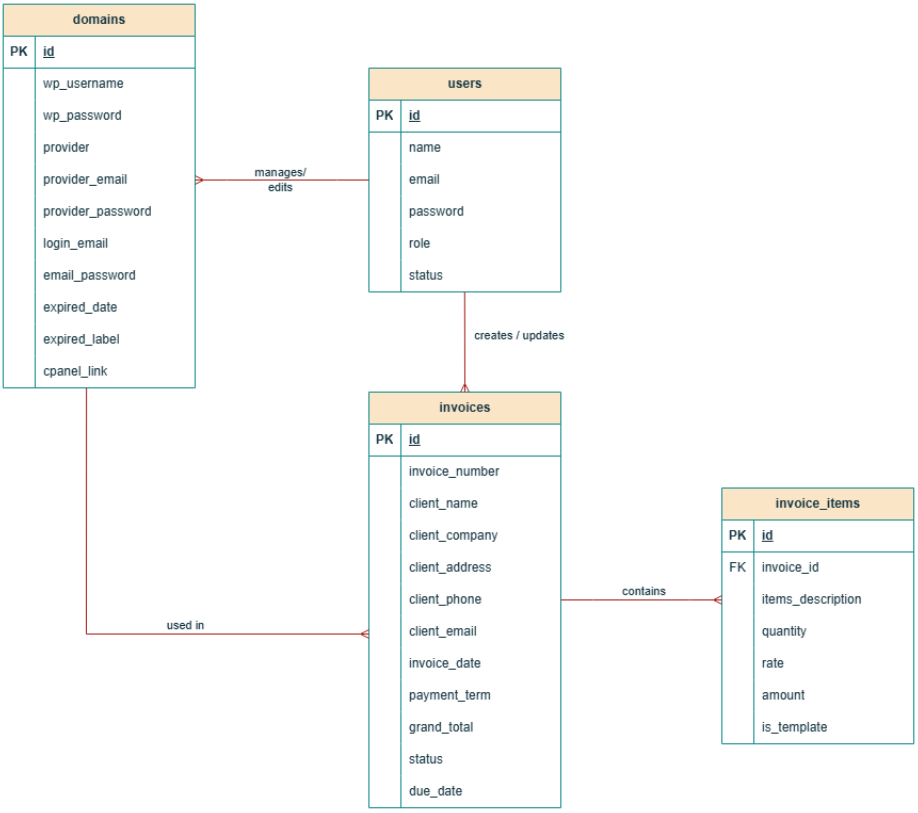


Figure 3.2: Entity Relationship Diagram (ERD) of DomTrack

Figure 3.2 consists of four core entities which is Users, Domains, Invoices, and Invoice Items. Each entity was created to support a specific part of the system's workflow. The Users entity stores both Admin and Staff accounts, including details such as name, email, password, role, and status. This table acts as the foundation for role-based access control, where Admin users can access all modules while Staff users are limited to viewing the dashboard, viewing settings, and generating invoices.

The Domains entity stores all domain-related information previously kept in Excel files. This includes the domain name, provider details, login credentials, expiration dates, and cPanel URLs. This table functions as the core dataset for reporting, so other modules especially the Reporting module, retrieve information directly from it. Because of this, a separate Report entity is not necessary.

Meanwhile, the Invoices and Invoice Items entities support the invoice generation process. Their relationship is designed as a one-to-many structure where a single invoice can contain multiple line items. The Invoices table stores general invoice details such as client name, contact information, dates, totals, and status. The Invoice Items table records each individual

item, including the description, rate, quantity, and calculated amount. These two tables are linked through the invoice_id field.

Additionally, the Users table is related to the Invoices table to indicate which user created or updated a particular invoice. This relationship supports accountability and ensures that every generated invoice can be traced back to the responsible user.

3.2.2 Wireframe Design

The design phase of DomTrack focused on creating a user-friendly interface and visualizing the workflow between the modules. Wireframes were created to provide a clear view of the system layout, showing how users interact with each module. Four key wireframes were selected to represent the main functionalities of the system. These wireframes illustrate the essential features while maintaining simplicity for review purposes. Additional wireframes exist and can be found in the appendix.

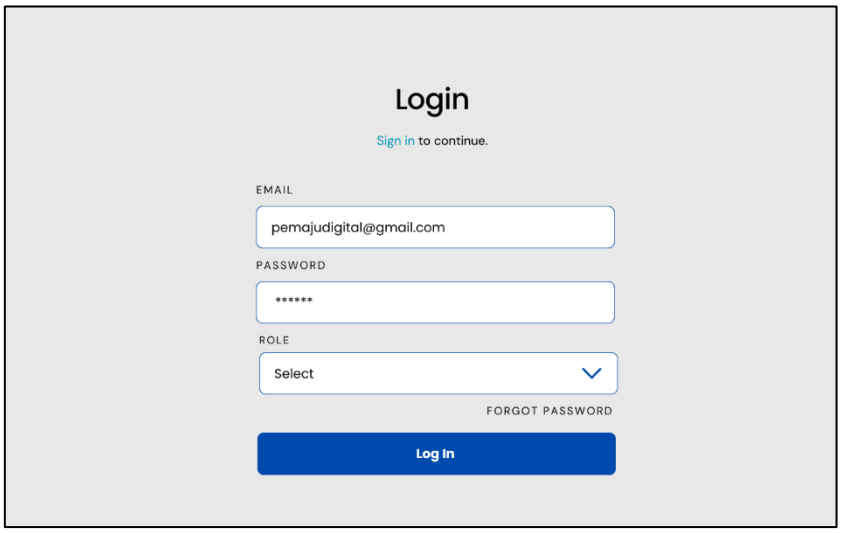


Figure 3.3: Login Page

Figure 3.3 shows login page that allows users to access the system according to their role. Both Administrators and Staff members enter their credentials, and the system verifies them before granting access. The wireframe shows fields for email and password, a login button, and a role-specific redirection upon successful authentication.

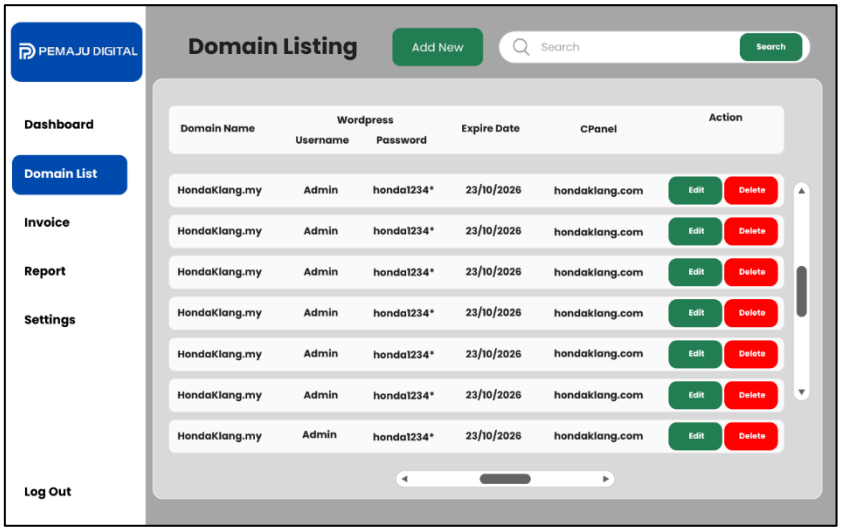


Figure 3.4: Domain Management Page

Figure 3.4 represents the core module for managing domain records. Administrators can add new entries, edit existing domains, and view all records in a structured table. Forms are connected to the database for real-time updates. The interface also includes search functionality to quickly locate specific domain entries. This module forms the central data source for reporting and invoice generation.

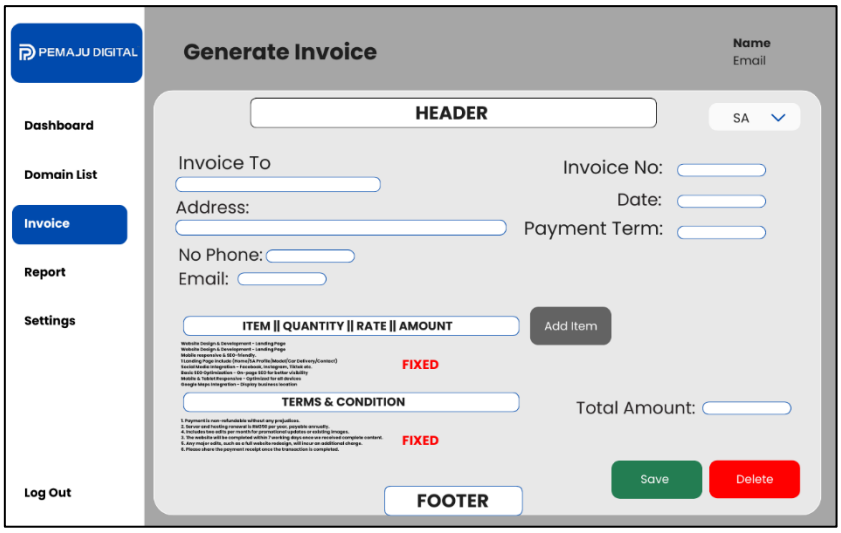


Figure 3.5: Generate Invoice Page

Figure 3.5 which is generate invoice page allows authorized users to generate invoices based on domain information. Staff members can create invoices, and the system automatically formats them for PDF export. Administrators have full access to manage invoice records, edit

details, and approve the final version. The wireframe shows fields for client details, item descriptions, and a generate button.

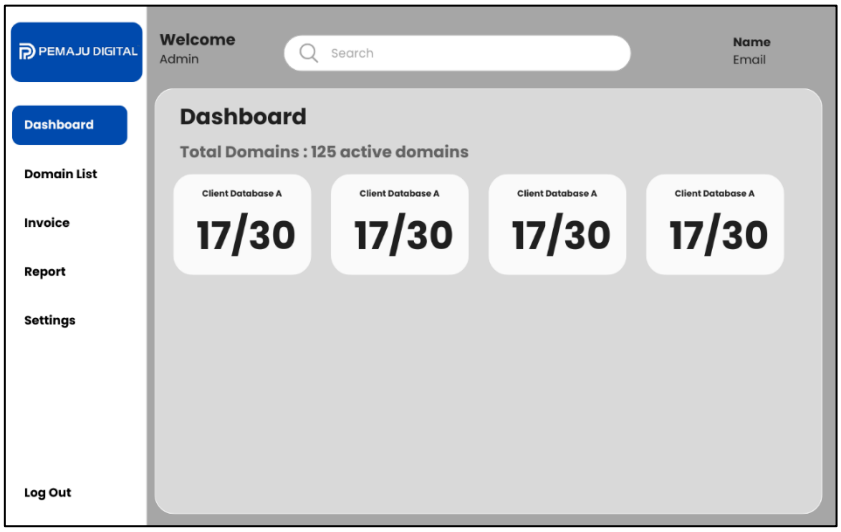


Figure 3.6: Dashboard Page

Figure 3.6 dashboard wireframe displays an overview of system activity. Administrators can view summaries of domains, invoices, and reports. Staff members can only see modules allowed for their role. The design emphasizes quick access to key metrics and highlights important actions for both user types.

3.2.3 User Interface Design

The DomTrack system interface was designed to support daily work in a simple and clear way. The main goal of the interface is to make the system easy to use for both Admin and Staff without requiring technical knowledge. Each page uses a consistent layout, so users do not need to relearn the system when moving between modules.

The layout of the system is divided into three main parts which is a header at the top, a sidebar menu on the left, and a content area in the center. The header displays basic system information, while the sidebar shows the available modules based on the user role. The content area is used to display forms, tables, and system information. This structure helps users understand where to find features and reduces confusion during system use.

Role-based access is applied to control system visibility and functions. Admin users are allowed to access all system modules, including domain management, invoice handling, report generation, system settings, and staff account management. Staff users are given limited access

and can only view the dashboard, view system settings, and generate invoices. This separation ensures that important system functions are handled by authorized users only.

The interface design focuses on clarity rather than visual complexity. Simple colors, clear text, and standard buttons are used across the system. Forms are arranged in a logical order, and tables are designed to show information in a readable format. This approach helps users complete tasks faster and reduces the chance of errors during data entry or system navigation.

a) Login Interface

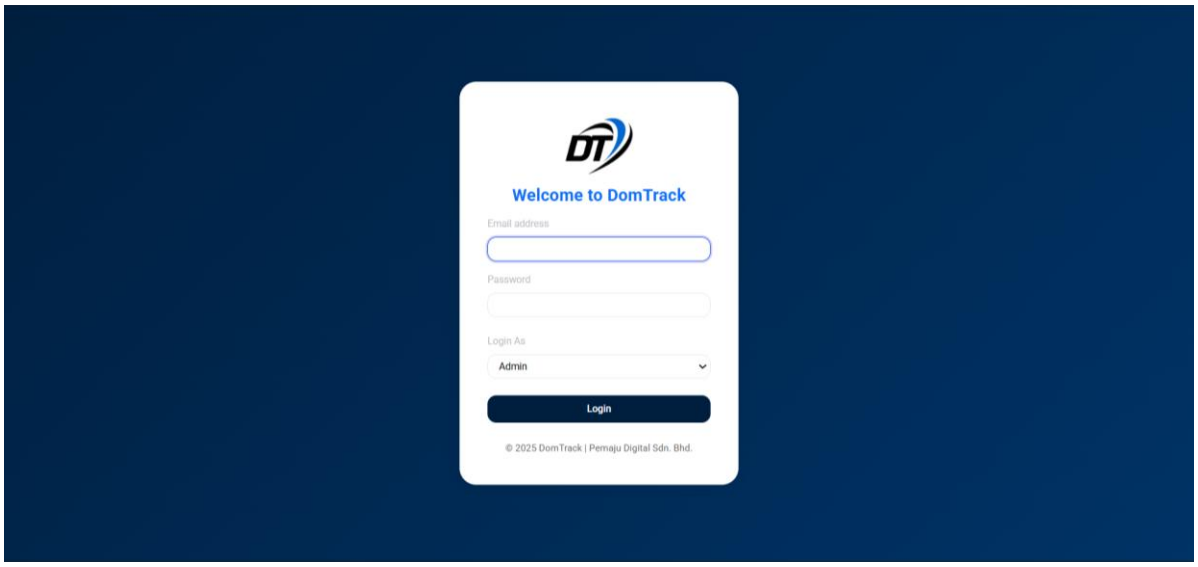


Figure 3.7: Login Interface

Figure 3.7 is the first page users see when accessing DomTrack. It is used by both Admin and Staff to enter the system securely. The page requires users to input their email and password and select their role from a dropdown menu before clicking the login button. The role selection ensures the system identifies the user type and grants access to the correct set of features.

Once the login button is clicked, the system validates the credentials against the database. If the credentials are incorrect, an error message appears and the user must try again. This prevents unauthorized access and ensures that only registered users can proceed.

After successful login, users are redirected to the dashboard according to their role. Admin gains full access to modules like domain management, invoices, reports, and staff accounts, while Staff are limited to features such as viewing the dashboard, settings, and

generating invoices. This ensures security and proper workflow management while keeping the interface simple and user-friendly.

b) Dashboard Interface

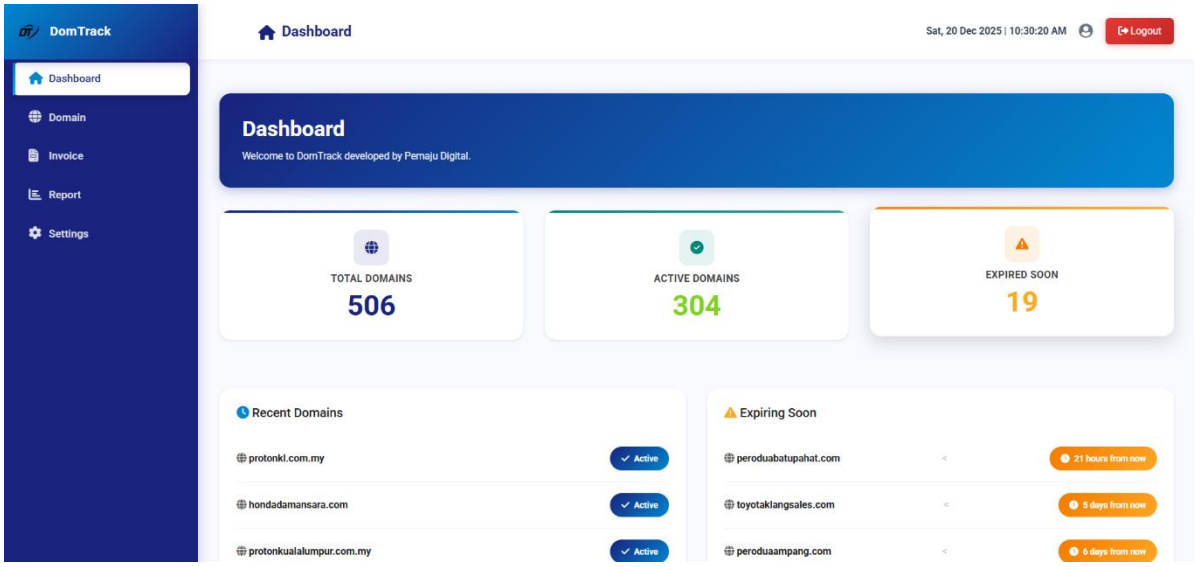


Figure 3.8: Dashboard Interface

Figure 3.8 shows the dashboard serves as the main page for users after logging in. It provides a quick overview of the company's domain information. Users can see the total number of domains, how many are active, which domains are expiring soon, recently added domains, and a separate list highlighting domains that will expire soon. Both Admin and Staff can view this page, making it the central point for monitoring system data.

The layout is designed to be simple and easy to understand. Key information is presented in cards at the top, while lists and tables are used to display detailed data. The visual design uses clear labels, minimal colors, and responsive tables so users can check information quickly on different devices.

The dashboard also helps users prioritize their tasks. Staff can check the status of domains and keep track of expiring ones. Admins get the same view but also have options to navigate to the domain management page if they need to update records. By having all important information in one place, the dashboard reduces the need to click through multiple pages and makes daily monitoring more efficient.

c) Domain Management Interface

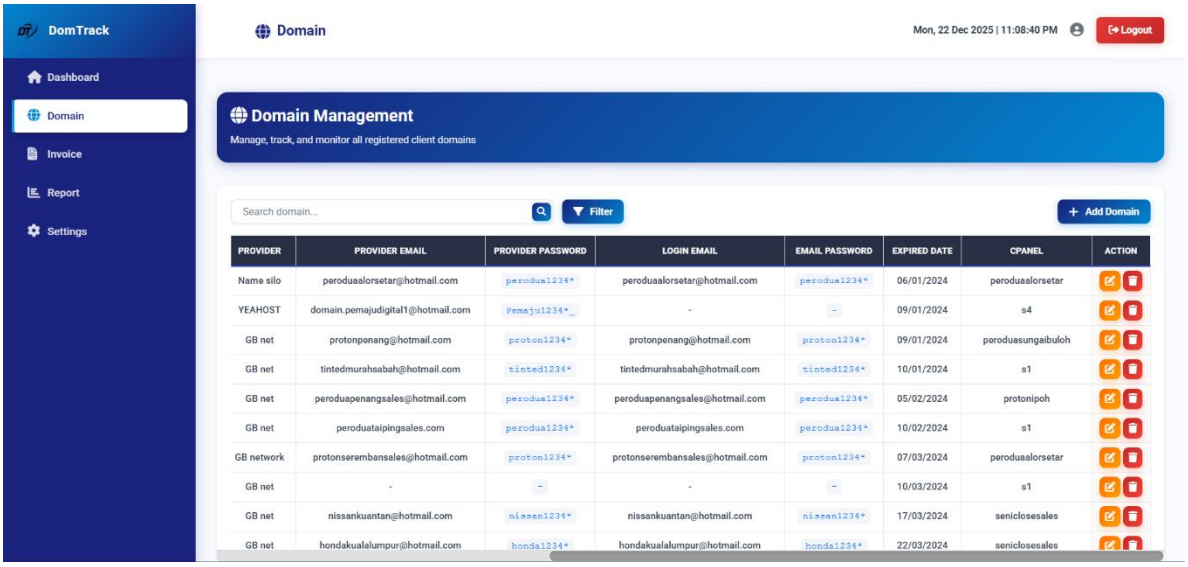


Figure 3.9: Domain List Interface

Figure 3.9 shows the domain management interface allows Admin users to view and manage all domain records in the system. The main screen displays a domain list containing information such as domain name, WordPress username and password, provider details, login email and password, expiry date, cPanel link, and available actions (edit and delete). This layout helps Admin users monitor domain information in a clear and organized manner.

A search and filter feature are provided at the top of the domain list to make data management easier. Admin users can search domains by name and apply filters to view active domains, expired domains, or domains with cPanel access. These filters help users quickly locate specific records, especially when managing many domains. An Add Domain button is also available on this page. When selected, the system navigates the user to the add domain form, where new domain information can be entered in a structured format that matches the domain list.

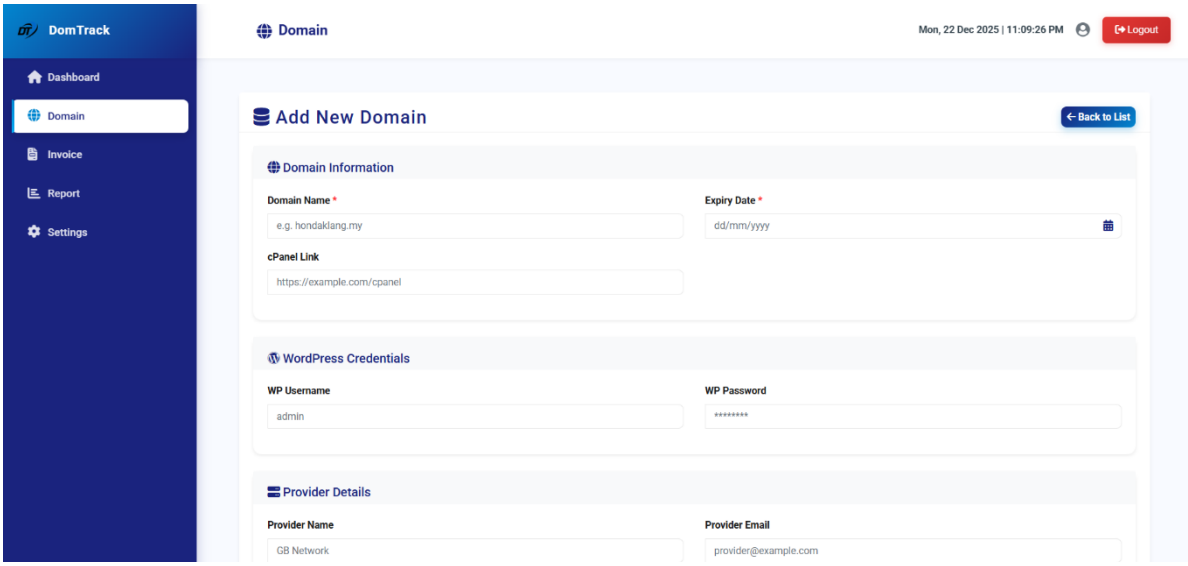


Figure 3.10: Add Domain Form Interface

The add domain form allows Admin users to insert new domain records into the system. The form fields follow the same structure as the domain list, including domain name, WordPress credentials, provider details, login information, expiry date, and cPanel link. This consistency ensures accurate data entry and reduces input errors.

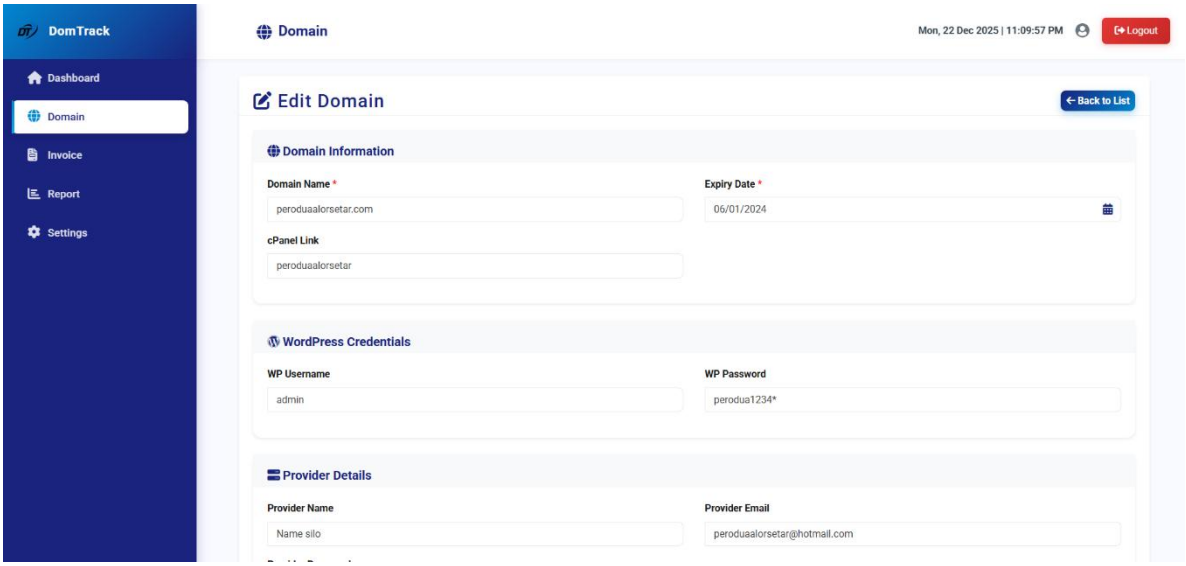


Figure 3.11: Edit Domain Form Interface

The edit domain form is used to update existing domain records. All current domain details are automatically displayed in the form, allowing Admin users to modify only the required fields. After saving, the updated information is reflected in the domain list, ensuring the data remains up to date and accurate.

d) Invoice Management Interface

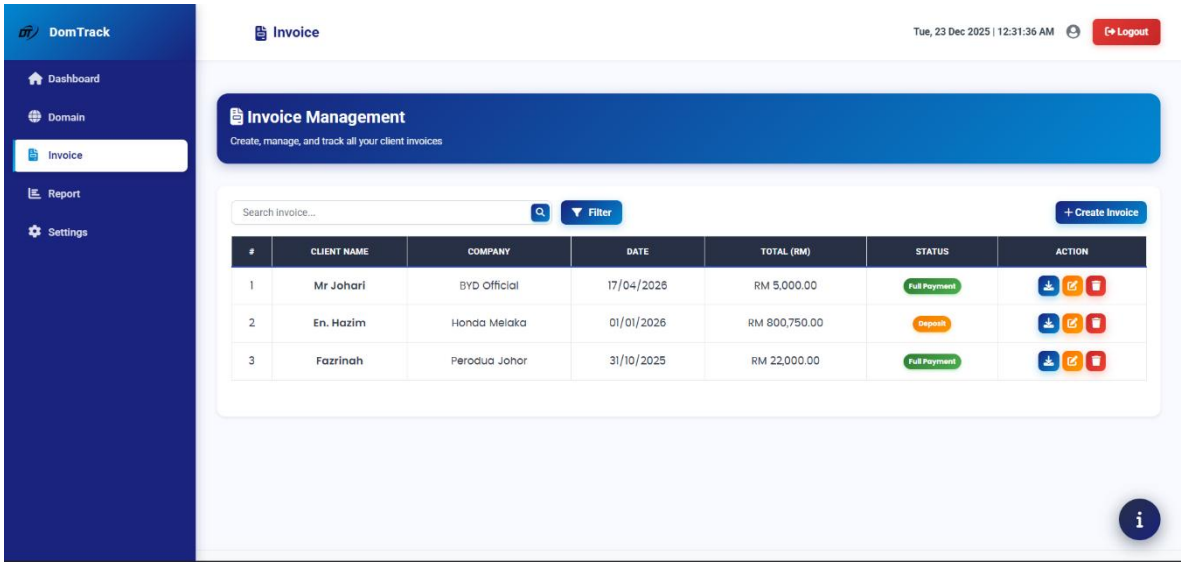


Figure 3.12: Invoice List Interface

The invoice list interface displays all invoices stored in the system. Each record shows the client name, company, invoice date, total amount (RM), and payment status. An action column is provided to allow users to download, edit, or delete an invoice. This interface helps users manage invoice records and monitor payment information efficiently.

The Download button allows admin and staff to generate and download the invoice in PDF format. The generated invoice follows a standard template that includes client details, invoice number, item list, total amount, payment terms, and company information. This feature enables users to produce a professional invoice directly from the system without manual formatting.

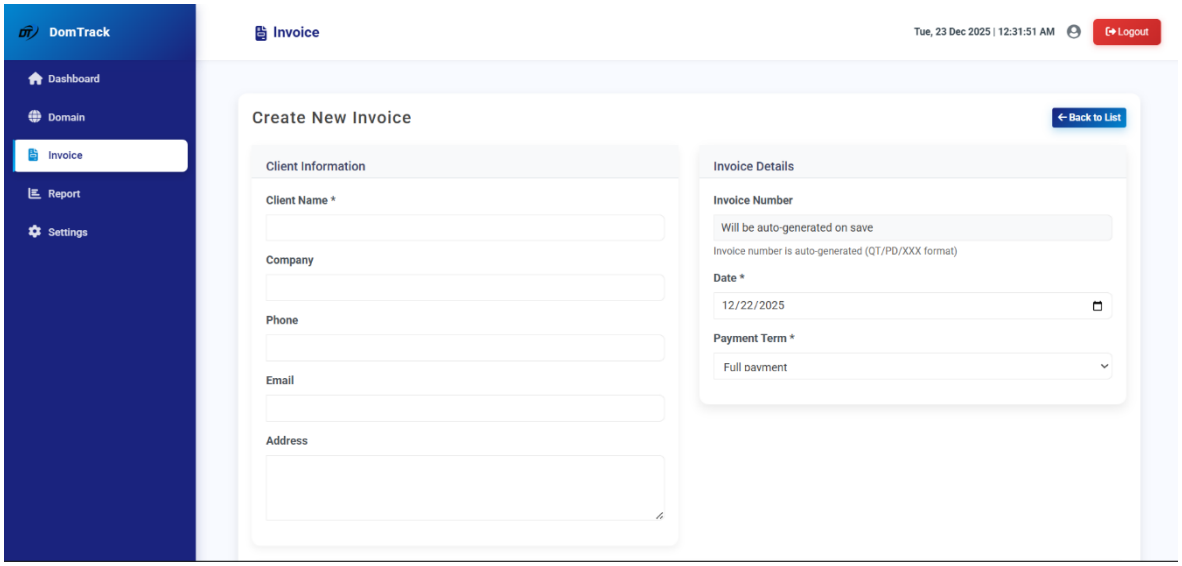


Figure 3.13: Create New Invoice Form Interface

The create invoice form is used to generate a new invoice. The form is divided into two main sections which is Client Information and Invoice Details. Client information includes fields such as client name, company, phone number, email, and address. Invoice details include invoice number, date, and payment term. The invoice number is automatically generated by the system when the invoice is saved, following the QT/PD/XXX format. This ensures that all invoices have a unique and consistent numbering structure. Required fields are clearly marked to prevent incomplete data submission.

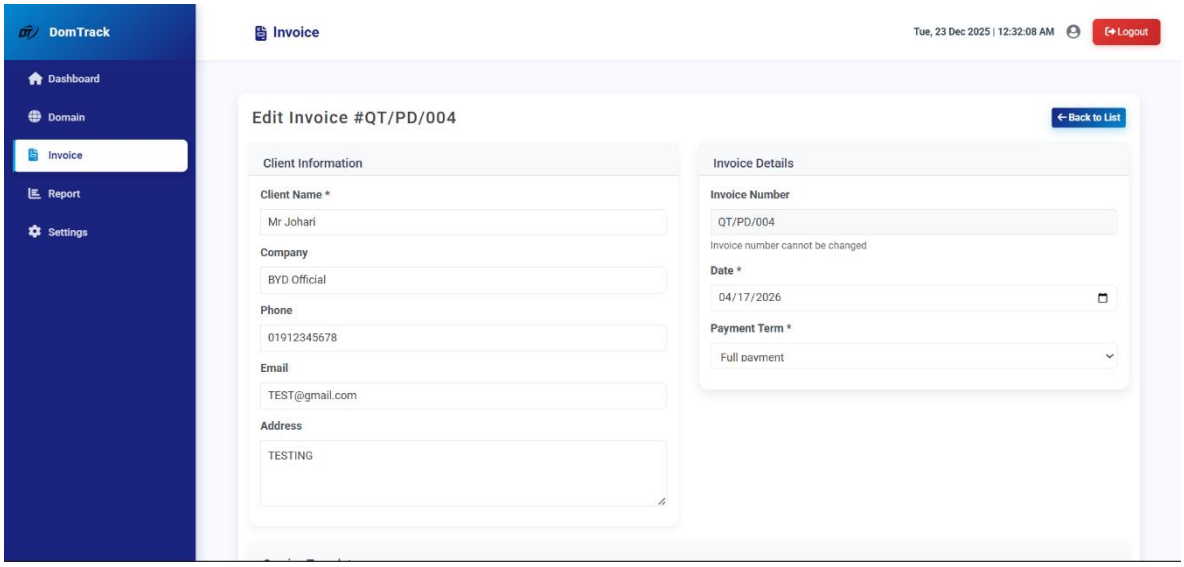
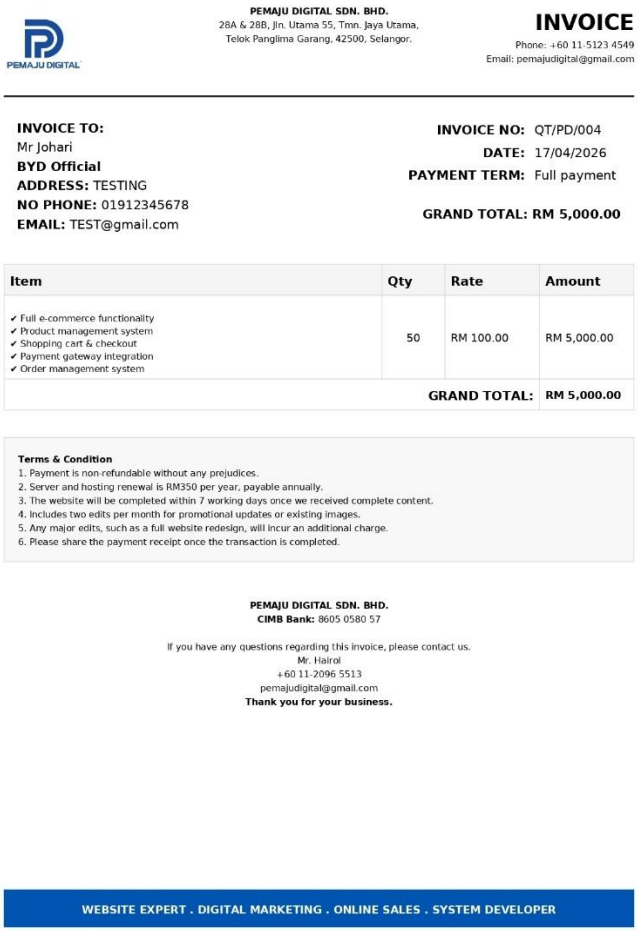


Figure 3.14: Edit Invoice Form Interface

The edit invoice form has the same layout and structure as the create invoice form. Existing invoice data is automatically displayed in the form fields, allowing users to update information such as client details, payment terms, or invoice date. Once saved, the updated information is reflected in the invoice list.



PEMAJU DIGITAL

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28A & 28B, Jln. Utama 55, Tmn. Jaya Utama,
Telok Panglima Garang, 42500, Selangor.

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Email: pemajudigital@gmail.com

INVOICE TO:
Mr Johari
BYD Official
ADDRESS: TESTING
NO PHONE: 01912345678
EMAIL: TEST@gmail.com

INVOICE NO: QT/PD/004
DATE: 17/04/2026
PAYMENT TERM: Full payment

GRAND TOTAL: RM 5,000.00

Item	Qty	Rate	Amount
✔ Full e-commerce functionality ✔ Product management system ✔ Shopping cart & checkout ✔ Payment gateway integration ✔ Order management system	50	RM 100.00	RM 5,000.00
		GRAND TOTAL:	**RM 5,000.00**

Terms & Condition
1. Payment is non-refundable without any prejudices.
2. Server and hosting renewal is RM350 per year, payable annually.
3. The website will be completed within 7 working days once we received complete content.
4. Includes two edits per month for promotional updates or existing images.
5. Any major edits, such as a full website redesign, will incur an additional charge.
6. Please share the payment receipt once the transaction is completed.

PEMAJU DIGITAL SDN. BHD.
CIMB Bank: 8605 0580 57

If you have any questions regarding this invoice, please contact us.
Mr. Hairol
+60 11-2096 5513
pemajudigital@gmail.com
Thank you for your business.

WEBSITE EXPERT . DIGITAL MARKETING . ONLINE SALES . SYSTEM DEVELOPER

Figure 3.15: Invoice Template

The figure shows the invoice template generated by the DomTrack system in PDF format after an invoice has been successfully created. The invoice contains client details such as name, company, address, phone number, and email, which are automatically retrieved from the system database. It also lists the services provided, including system features delivered to the client, allowing clear understanding of the scope of work. This structured layout helps ensure accuracy and professionalism in billing documentation.

In addition, the invoice includes important transaction information such as invoice number, issue date, payment terms, quantity, rate, and the calculated grand total amount. The Terms and Conditions section outlines payment policies, project timeline, and service limitations to avoid misunderstandings. Company and bank details are provided to facilitate payment, along with official contact information for inquiries. Overall, this invoice template improves efficiency, transparency, and consistency in the invoicing process.

e) Report Management Interface

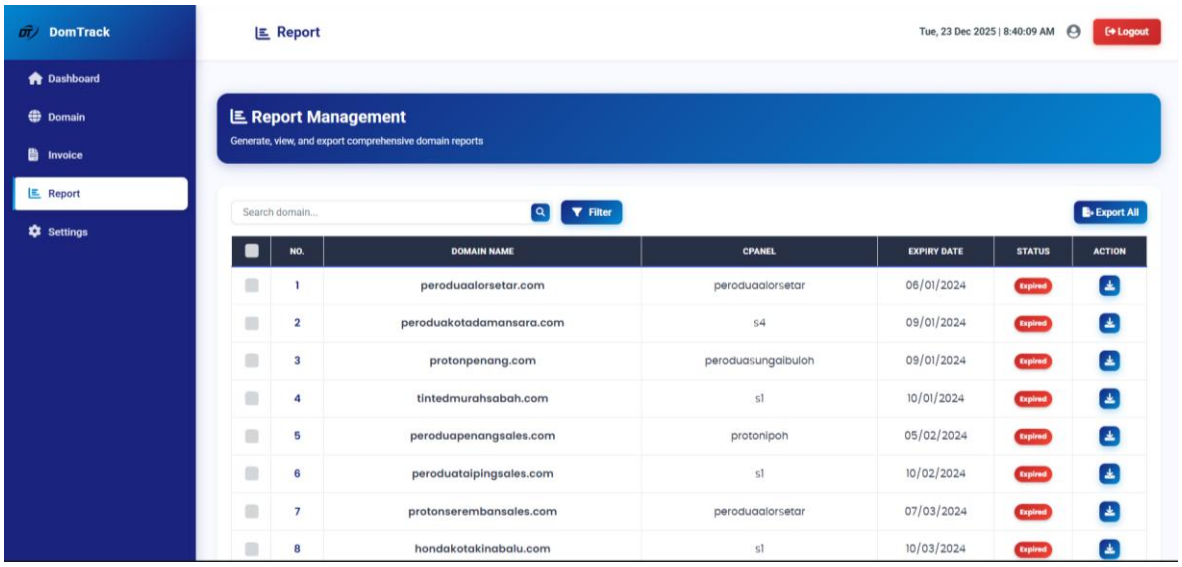

Figure 3.16: Report Management Interface

The report interface is used to view and generate domain summary reports. It displays domain information in a table format to make the data easy to read and manage. Each record shows the domain name, cPanel status, expiry date, and domain status (expired or not expired).

A search function is provided to allow users to quickly find specific domains. Admin can also filter the data based on domain status to view expired or active domains only. These functions help reduce the time needed to locate specific records, especially when handling a large number of domains.

The Export All Report function allows users to generate a complete report in PDF format. When this option is selected, the system retrieves all domain records from the database and exports them into a structured report file. In addition, a Download button is available for each record in the table, allowing users to generate and download a report for an individual domain.

f) Setting Interface

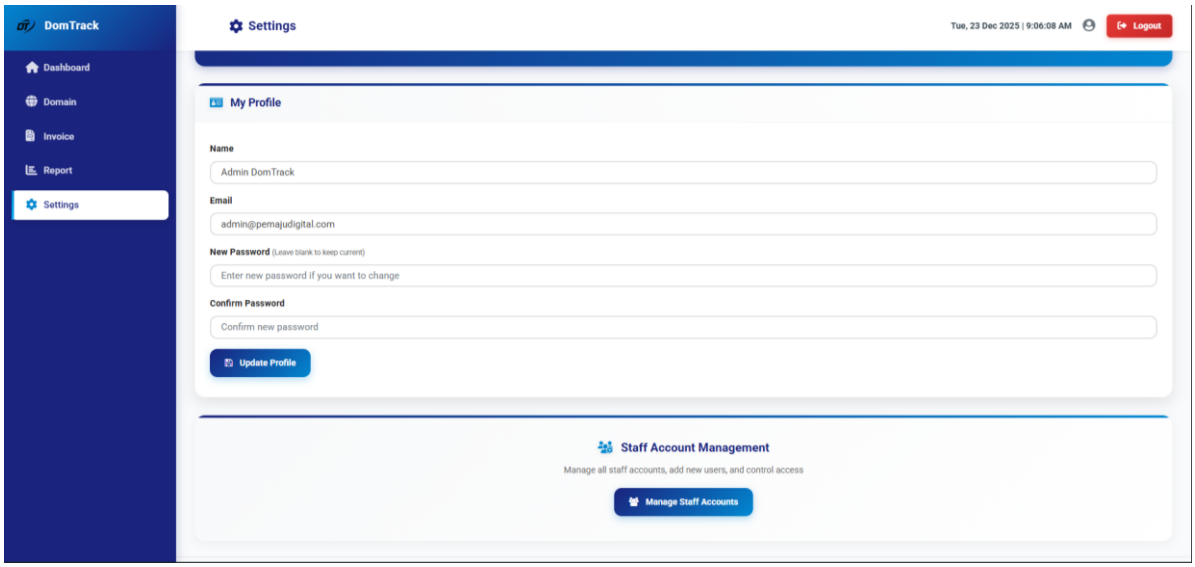

Figure 3.17: Admin Setting Interface

The setting interface allows the admin to manage personal profile information and system user access. In this interface, the admin can update profile details such as name, email, and password. Any changes made will be saved and applied immediately to the account. Below the profile section, a button is provided to allow the admin to access the staff account management page. This function is restricted to admin users only to ensure proper control over system access.

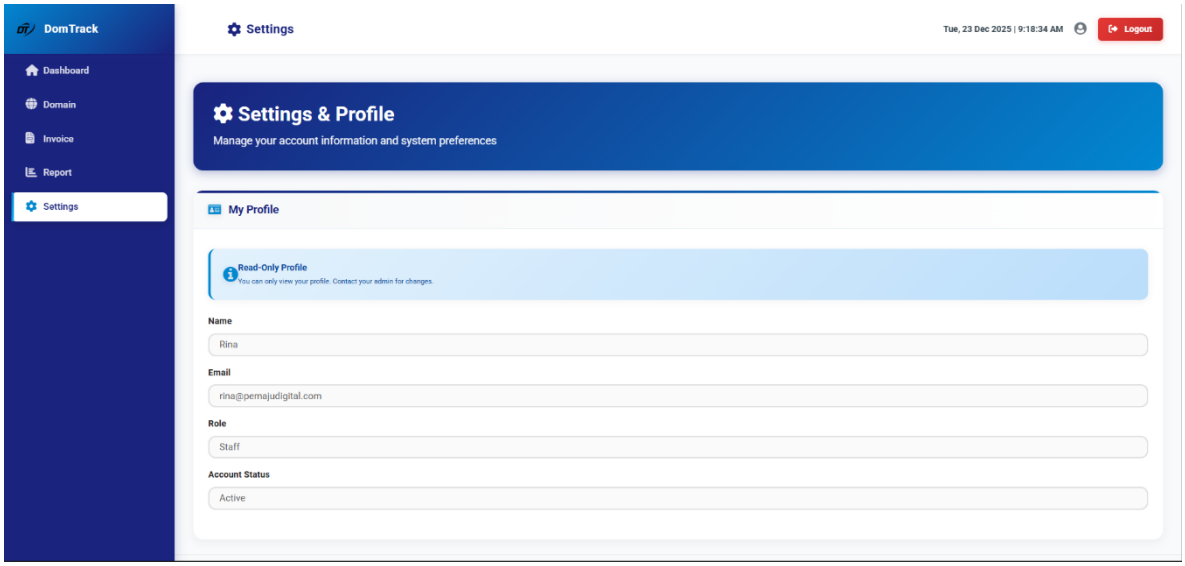

Figure 3.18: Staff setting Interface

The staff setting interface displays the staff user's profile information in a read-only format. This interface allows staff members to view their personal details such as name, email, role, and account status, but editing is not permitted.

This restriction ensures that staff users cannot modify their own account information or system-related settings. Any updates to staff profiles must be performed by the admin through the account management interface to maintain data accuracy and system security.

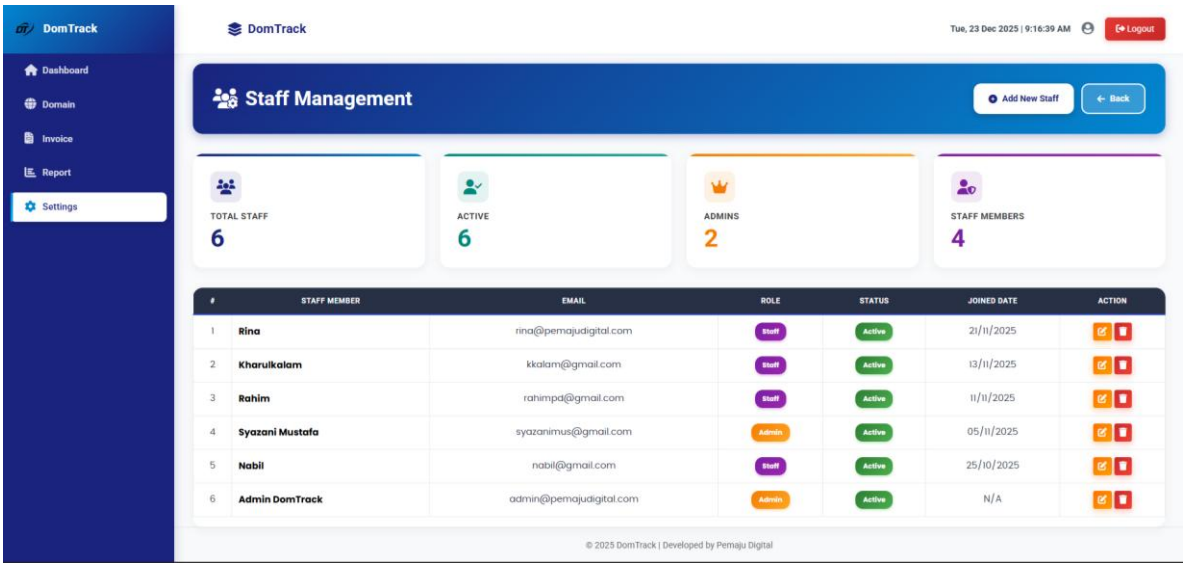

Figure 3.19: Staff Management Interface

The account management interface displays a table containing the list of staff who are allowed to access the system. The table shows staff information including name, role, account status, joined date, and available actions. The admin can manage staff accounts by using the Edit and Delete actions provided in the table. An Add New Staff button is available to create new staff accounts and assign appropriate roles. At the top of the page, a summary section displays the total number of staff accounts, the number of active users, and the breakdown between admin and staff members. This summary helps the admin monitor user access and manage system roles more effectively.

3.2.4 Process Flow Design

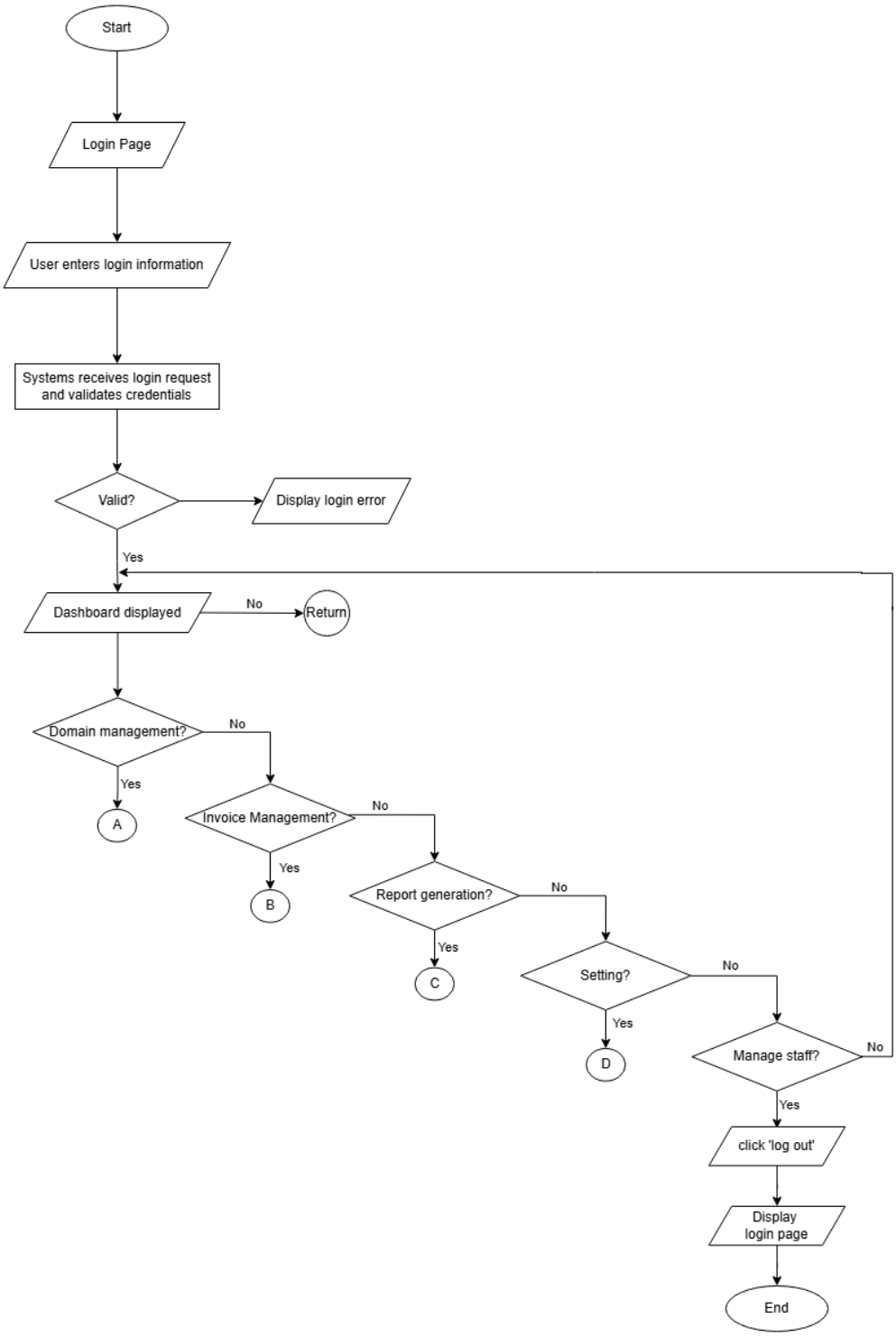

Figure 3.20: Flowchart admin module

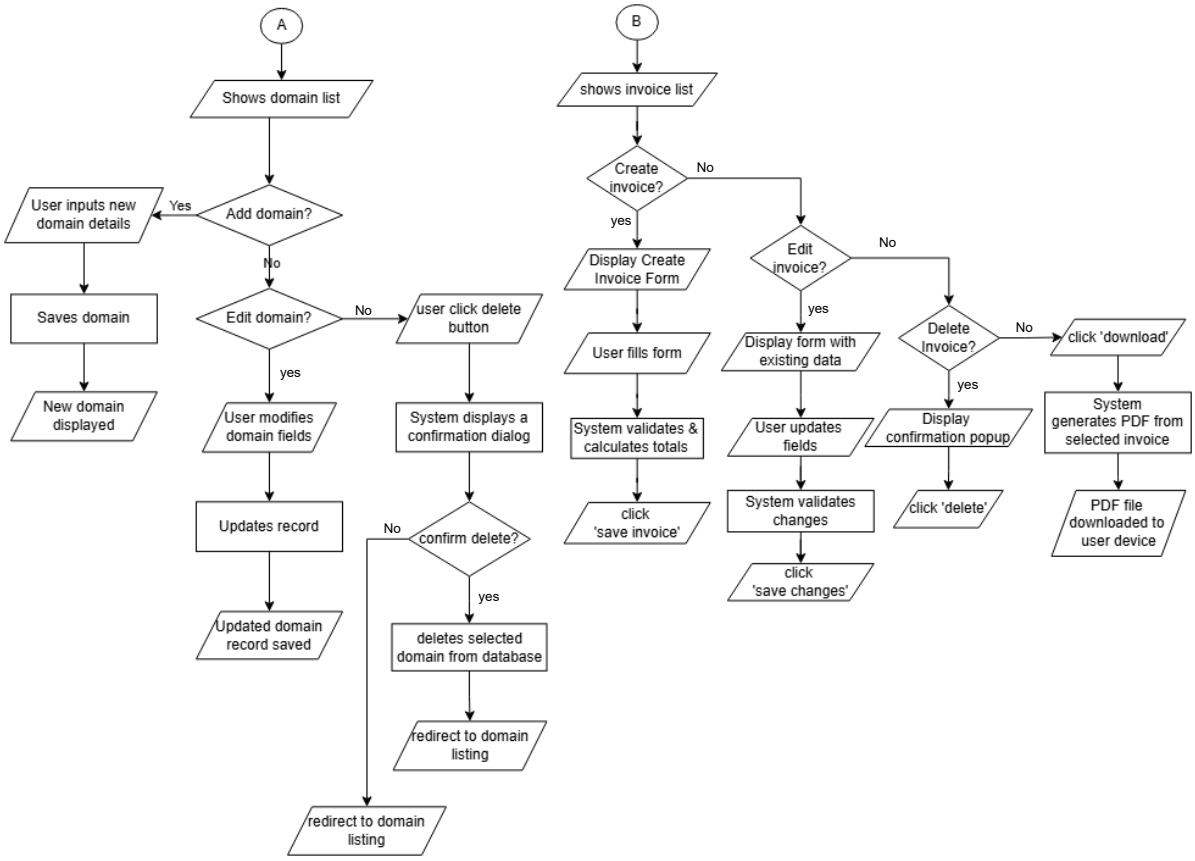

Figure 3.20: Flowchart admin module (continue)

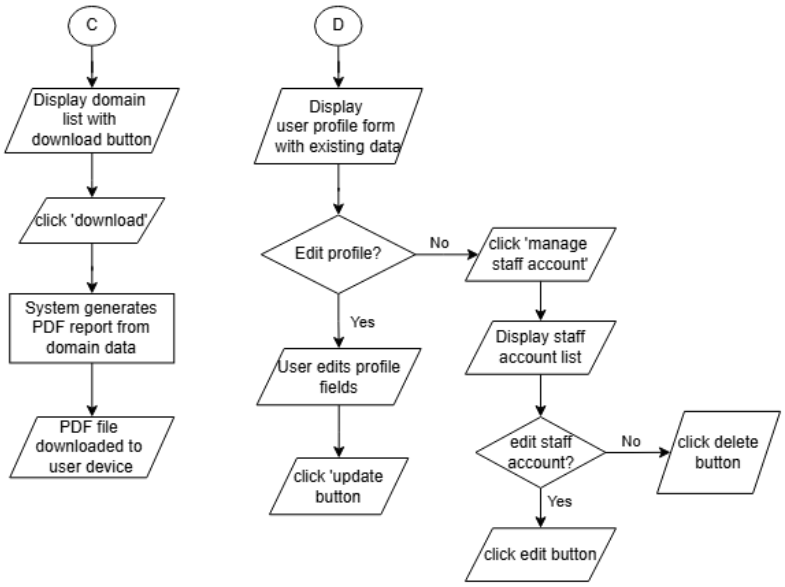

Figure 3.20: Flowchart admin module (continue)

Figure 3.7 shows the flowchart for the Admin Module illustrates the entire sequence of interactions performed by an administrator when using the DomTrack system. The process begins with the login stage, where the admin is required to enter valid credentials before gaining access to the system. If the information provided is incorrect, the system will return an error and prompt the user to reattempt the login. Once the credentials are validated, the admin is directed to the dashboard, which acts as the central navigation point for all system functions.

From the dashboard, the admin can access several key modules, including domain management, invoice handling, report generation, system settings, and staff account management. Each module follows a distinct internal process. For example, within the Domain Management module, the administrator can view the full domain listing and choose to add, edit, or delete records. Adding or editing a domain requires the admin to fill in or update domain details, after which the system validates and stores the information before returning to the listing. Deleting a domain requires a confirmation step to prevent accidental removal, and once confirmed, the record is permanently removed from the database.

The Invoice Management module similarly begins with an invoice listing, from which the admin may create a new invoice, modify an existing one, or delete unwanted entries. The system also allows the admin to generate downloadable PDF invoices automatically. For report generation, the flow is more direct upon selecting the report option, the system retrieves the domain data and immediately generates a PDF file for download. Settings and staff management are also accessible from the dashboard, enabling the admin to update profile details or manage staff accounts through edit or delete actions. The flow ends when the administrator chooses to log out, which terminates the session and returns the user to the login page.

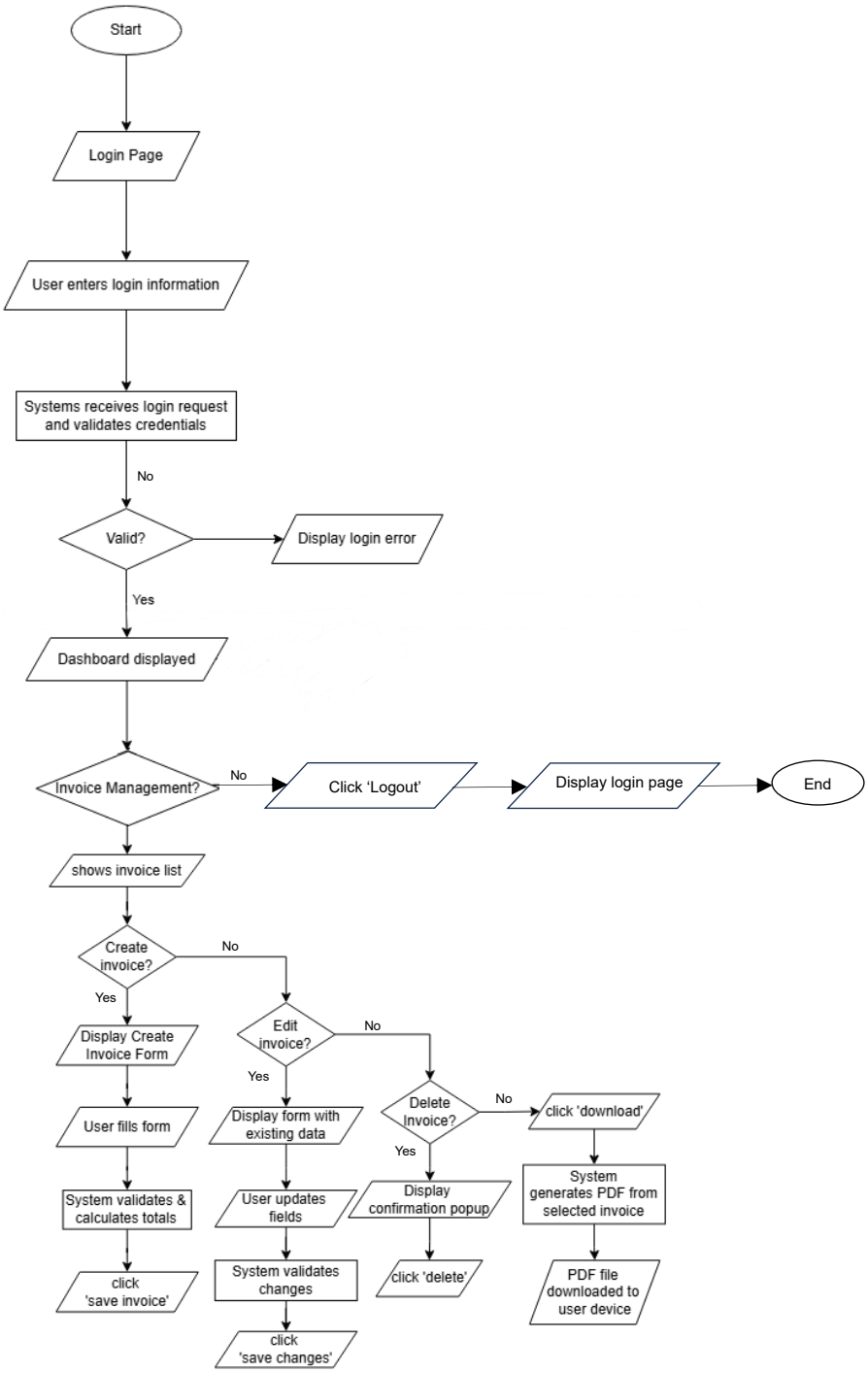

Figure 3.21: Flowchart staff module

Figure 2.9 shows the flowchart for the Staff Module outlines a simplified process flow that reflects the limited and controlled access provided to staff users in the DomTrack system. The sequence begins at the login page, where staff members are required to enter their credentials. The system validates the information submitted, ensuring that only authorised users can proceed. If the credentials are incorrect, the system returns an error message and prompts the

user to try again. Upon successful login, the staff member is redirected to the dashboard, which serves as the main interface for all available staff-level functions.

From the dashboard, the staff member can view system information and navigate to the invoice module. The invoice generation feature allows staff to create invoices based on the stored domain records. When selecting this module, the system displays the invoice form where relevant customer and billing details can be filled in. After the required information is provided, the system validates the inputs, generates the corresponding invoice, and stores it in the system. Staff members may then download the invoice in PDF format, enabling them to provide customers with properly formatted billing documents.

Once staff have completed their tasks, they can return to the dashboard and choose to log out. Selecting the logout option immediately terminates the session and returns the user to the login page, ensuring that system access remains secure. The overall flow reflects a straightforward, task-focused interaction designed to give staff only the functions necessary for their daily operations without exposing administrative features.

3.3 Implementation Phase

The implementation phase involves developing the DomTrack system based on the design specifications defined earlier. This stage focuses on building the backend system using Laravel, designing the user interface with Blade and Bootstrap, and integrating all components with a MySQL database. The environment setup was performed using Laravel version 12.32.5 with PHP 8.2.12, supported by a MySQL database named domain_db. Blade templates were used to structure the views, while Bootstrap provided a clean and responsive visual layout. The project was initially developed in a local environment before being prepared for deployment. During setup, environment variables were configured, database migrations were executed, and essential packages such as Laravel Breeze (for authentication) and DomPDF (for PDF generation) were installed.

Backend development covered the implementation of several core modules. The authentication module, created using Laravel Breeze, manages login, logout, session handling, and access control for both Admin and Staff roles. Middleware was applied to ensure that each user type can only access the features assigned to their role. The domain management module, which forms one of the main components of DomTrack, was developed to support operations

such as adding new domains, editing existing entries, deleting records, and displaying a complete domain list. The dashboard module displays key summaries such as the total number of domains, upcoming expiries, and general system activity. The invoice module allows the creation, editing, deletion, and PDF export of invoices, while the report module enables administrators to download domain summary reports in PDF format.

The frontend implementation uses Blade templating to deliver structured, reusable layouts across the system, including the header, sidebar navigation, and content sections. Bootstrap was applied to ensure responsive design, consistent spacing, and user-friendly forms and tables. The interface was developed to be simple, clean, and easy for non-technical staff to navigate. Role-based views ensure that staff account holders only access the dashboard and invoice features, while administrators have full access to additional modules such as domain management, reports, and staff account settings.

Database implementation was managed using Laravel migrations, which define the structure of tables such as users, domains, invoices, and invoice items. Laravel models were used to handle database operations and manage interactions between system components and the MySQL database. This approach allows data to be created, updated, and retrieved in an organised way, while supporting relationships between related records such as invoices and their items.

Throughout the implementation stage, the system was continuously tested to ensure all modules worked as intended. Feature testing, validation checks, and manual testing were carried out for processes such as login authentication, CRUD operations, PDF generation, and navigation flow. Any bugs such as UI misalignment, incorrect queries, or validation errors were resolved during development. This iterative approach ensured that the DomTrack system performed smoothly and reliably before progressing to the testing and deployment phases.

3.4 Testing Phase

The testing phase aimed to ensure that all modules of the DomTrack system functioned correctly according to the project objectives. Each module was tested individually to verify its behavior, input validation, data storage, and output generation. Both functional and usability aspects were checked to ensure smooth user experience for Admin and Staff roles.

Before testing, a test plan was prepared, which included defining test scenarios, expected results, and criteria for passing or failing. Testing involves performing real tasks, such as logging in as Admin or Staff, adding and updating domain entries, generating invoices, and exporting reports to PDF. Observations were recorded for comparison with expected outcomes.

Table 3.1 shows the summary of testing results for key modules in DomTrack. The table highlights the module, test scenario, expected result, actual outcome, and the pass/fail status for each test case. Placing the table here allows the reader to see the comprehensive testing coverage immediately after the introduction to the testing phase.

Table 3.1: Testing results

Module	Test Scenario	Expected Result	Actual Result	Status
Authentication	Admin login with correct credentials	Login successful, dashboard displayed	Login successful	Pass
Authentication	Staff login with wrong password	Error message shown	Error message shown	Pass
Domain Management	Add new domain entry	Entry saved and displayed in list	Entry saved and shown	Pass
Invoice Generation	Generate PDF invoice	PDF file created with correct data	PDF generated	Pass
Report Generation	Generate domain report PDF	Report downloaded with accurate info	Report generated	Pass

Table 2.4 clearly shows that all key functionalities of the DomTrack system passed the testing phase. This confirms that the system is reliable, functional, and ready for deployment in the company environment.

3.5 Deployment Phase

The deployment phase involves transferring the DomTrack system from the local development environment to the company's production server so that it can be accessed by authorized users. Prior to deployment, the server environment was prepared using cPanel to ensure compatibility with the system requirements. This included configuring the server with PHP version 8.2, creating a MySQL database, and enabling the required PHP extensions. cPanel was used to manage server settings, databases, and application files efficiently.

After the server environment was ready, the system files were uploaded from the local machine to the hosting server using the File Manager provided by cPanel. The environment configuration file was then updated to match the production settings, including database credentials, application URL, and security keys. Database migrations were executed through the server to create the required tables such as users, domains, invoices, and invoice items. This ensured that the database structure in the production environment matched the structure developed during the implementation phase.

Once deployment was completed, functional testing was carried out on the live system. This included verifying login access for both Admin and Staff roles, checking domain management operations, and testing invoice and report generation in PDF format. Role-based access control, data integrity, and page navigation were also reviewed to ensure system stability. After all modules were confirmed to be functioning correctly, the DomTrack system was made available for daily use. This deployment phase completed the transition from a manual Excel-based workflow to a fully operational web-based system.

3.6 Maintenance Phase

The maintenance phase focuses on ensuring that the DomTrack system continues to operate smoothly after deployment. Once the system is in active use, regular monitoring is required to detect issues such as system errors, data inconsistencies, or performance problems. Maintenance activities help ensure that the system remains reliable and supports the company's daily operations without disruption.

During this phase, corrective maintenance is carried out whenever problems are identified. This includes fixing bugs reported by users, correcting incorrect data handling, and resolving issues related to user access or system functionality. Updates may also be required

when changes occur in business processes, such as new domain information fields or adjustments to invoice formats. These updates are applied carefully to avoid affecting existing system data.

Preventive and adaptive maintenance are also considered to keep the system up to date. This involves monitoring server performance, updating system dependencies when necessary, and ensuring data backups are performed regularly through cPanel. As user requirements evolve, minor improvements can be introduced to enhance usability and efficiency. Through continuous maintenance, the DomTrack system is able to remain stable, secure, and relevant to the company's long-term operational needs.

3.7 Project Timeline

Student Name: Nurfazrina binti Azmi
Matric Number: AI220354
Company Name: Pemaju Digital SDN BHD

No.	Task	Start Date	End Date	Duration
1	**Setup environment (XAMPP/Laragon, Composer, Laravel) and initial database design**	**6/10/2025**	**11/10/2025**	6 Days
	a. Install and configure XAMPP/Laragon	6/10/2025	6/10/2025	
	b. Setup Composer and Laravel framework	7/10/2025	8/10/2025	
	c. Create project folder structure	9/10/2025	9/10/2025	
	d. Draft initial ERD for databas	10/10/2025	11/10/2025	
2	**Create Laravel migrations and build database tables (users, cPanels, domains)**	**13/10/2025**	**18/10/2025**	6 Days
	a. Write migration files for users, cPanels, domains	13/10/2025	14/10/2025	
	b. Define table fields and relationships	14/10/2025	15/10/2025	
	c. Run migrations in MySQL	15/10/2025	17/10/2025	
	d. Insert dummy test data	18/10/2025	18/10/2025	
3	**Develop login and authentication module (role-based access for admin/staff)**	**20/10/2025**	**25/10/2025**	6 Days
	a. Install Laravel Breeze/Jetstream	20/10/2025	20/10/2025	
	b. Create login & register forms	21/102025	22/102025	
	c. Apply role-based access (admin/staff)	22/102025	24/102025	
	d. Setup session & logout function	23/102025	25/102025	
4	**Continue login and authentication module (testing & refinement)**	**27/10/2025**	**1/11/2025**	6 Days
	a. Test login & register forms	27/102025	27/102025	
	b. Validate role permissions	28/102025	29/102025	
	c. Debug errors in authentication flow	30/102025	1/11/2025	
	d. Test create and update features	1/11/2025	1/11/2025	
5	**Implement CRUD functions for domain and cPanel records**	**3/11/2025**	**8/11/2025**	6 Days
	a. Build add/edit/delete forms	3/11/2025	4/11/2025	
	b. Validate inputs before saving	5/11/2025	6/11/2025	
	c. Connect CRUD to MySQL database	6/11/2025	8/11/2025	
6	**Continue CRUD module (testing & debugging)**	**10/11/2025**	**15/11/2025**	6 Days
	a. Test create and update features	10/11/2025	12/11/2025	
	b. Debug database errors	12/11/2025	13/11/2025	
	c. Refine input validation	14/11/2025	15/11/2025	
7	**Integrate CRUD with database and test data entry/update**	**17/11/2025**	**22/11/2025**	6 Days
	a. Link CRUD with cPanel-domain relationship	17/11/2025	18/11/2025	
	b. Test expiry date field updates	18/11/2025	19/11/2025	
	c. Debug errors in database transactions	20/11/2025	22/11/2025	
8	**Develop dashboard module to display cPanel statistics and active domains**	**24/11/2025**	**29/11/2025**	6 Days
	a. Create dashboard UI with Bootstrap	24/11/2025	26/11/2025	
	b. Display total domains per cPanel	25/11/2025	27/11/2025	
	c. Show quota usage (out of 100)	27/11/2025	27/11/2025	
	d. Add summary cards/counters	27/11/2025	29/11/2025	
9	**Add search function for quick domain lookup**	**1/12/2025**	**6/12/2025**	6 Days
	a. Add search bar in UI	1/12/2025	2/12/2025	
	b. Enable keyword search by domain name	2/12/2025	4/12/2025	
	c. Optimize search query for speed	5/12/2025	6/12/2025	
10	**Build PDF export feature for client reports**	**8/12/2025**	**13/12/2025**	6 Days
	a. Integrate Laravel DomPDF	8/12/2025	10/12/2025	
	b. Design report layout template	10/12/2025	11/12/2025	
	c. Include domain name & expiry date	10/12/2025	12/12/2025	
	d. Generate sample client report	12/12/2025	13/12/2025	
11	**System integration testing and bug fixing**	**15/12/2025**	**20/12/2025**	6 Days
	a. Combine all modules	15/12/2025	17/12/2025	
	b. Run system integration tests	17/12/2025	18/12/2025	
	c. Fix backend and UI issues	18/12/2025	20/12/2025	
12	**User acceptance testing with staff and final adjustments**	**22/12/2025**	**27/12/2025**	6 Days
	a. Present working system to staff	22/12/2025	22/12/2025	
	b. Gather feedback on usability	23/12/2025	24/12/2025	
	c. Apply UI/UX adjustments	24/12/2025	26/12/2025	
	d. Fix reported issues	26/12/2025	27/12/2025	
13	**Documentation, presentation, and deployment**	**29/12/2025**	**3/1/2026**	6 Days
	a. Prepare technical documentation	29/12/2025	30/12/2025	
	b. Write user manual	30/12/2025	30/12/2025	
	c. Finalize report and presentation slides	30/12/2025	1/1/2026	
	d. Deploy system on hosting/server	1/1/2026	3/1/2026	

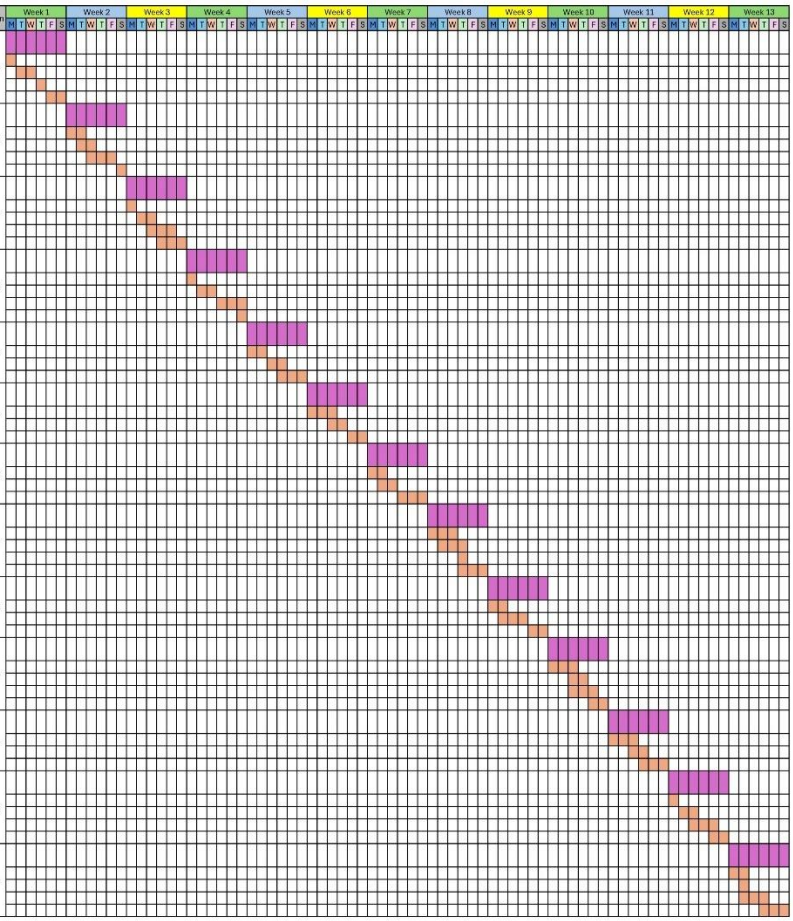

Figure 3.22: Gantt Chart Project

CHAPTER 4

DISCUSSION & ANALYSIS

4.1 Challenges Faced During Industrial Training

During the industrial training period, challenges were encountered while handling both daily operational tasks and a student-proposed system development project. Daily tasks mainly involved managing WordPress landing pages, learning and applying basic SEO practices, and assisting in the development of new automotive prospect systems for several car brands such as Honda, Perodua, Proton, and Toyota. In parallel, a larger-scale system development project, DomTrack, was developed using Laravel and MySQL.

One of the main challenges was adapting to different platforms and workflows. Daily website-related tasks were handled using WordPress, which required understanding theme customization, content arrangement, and responsiveness without affecting existing client websites. Learning SEO concepts was also challenging at the beginning, as it required understanding how content structure, keywords, and page performance influence search engine visibility.

Another technical challenge was working on multiple systems simultaneously. The automotive prospect systems were developed in a local environment using Laragon, which involved handling form submissions, data storage, and system flow for different brands. Ensuring that each system followed correct logic and data structure required careful testing and attention to detail.

For the DomTrack project, challenges included implementing backend logic using Laravel, managing database relationships through phpMyAdmin, and ensuring that the system met functional requirements such as role-based access control and data accuracy. Understanding how to structure a complete system from requirements to implementation required continuous learning and problem-solving.

From a non-technical perspective, time management was a significant challenge. Balancing daily tasks with ongoing system development work required proper task prioritization. Communication with supervisors and team members was also essential to ensure that requirements and expectations were clearly understood.

4.2 Solutions Implemented and Approaches Used

To overcome these challenges, structured approaches were applied throughout the training period. For WordPress-related tasks, existing templates and plugins were used effectively to build landing pages while maintaining consistency and responsiveness. SEO implementation was carried out gradually by applying fundamental practices such as proper heading usage, keyword placement, and content organization, based on guidance from supervisors and reference materials.

For the automotive prospect systems developed using Laragon, systematic testing was performed to ensure that data input, storage, and retrieval worked correctly. Each system was tested individually to confirm that the prospect data for different car brands was handled accurately.

In the DomTrack project, Laravel's MVC structure was utilized to organize system logic clearly. Database tables were managed using phpMyAdmin, and Git was used to track development progress and manage code changes. Regular testing and debugging were carried out to identify errors early and ensure system stability. Feedback from supervisors was also taken into account to improve system functionality and usability.

Non-technical challenges were addressed by improving communication and planning. Tasks were clarified through discussions, and deadlines were managed by dividing work into smaller, manageable phases.

4.3 Skills and Competencies Gained

The industrial training experience provided significant improvement in both technical and professional skills. Technical skills were enhanced through exposure to different development platforms. Working with WordPress improved understanding of website structure, landing page design, and content management. Learning SEO increased awareness of how websites can be optimized to improve visibility and user engagement.

System development tasks strengthened backend development skills, particularly through the DomTrack project using Laravel and MySQL. Practical experience was gained in database design, CRUD operations, authentication handling, and role-based access control. Experience using Laragon for local system development improved understanding of development environments and system testing. Git usage enhanced version control practices and collaborative development skills.

In terms of soft skills, the training improved time management, adaptability, and communication skills. Working in a real company environment requires responsibility, discipline, and the ability to adapt to different tasks and technologies within a limited timeframe.

4.4 Lessons Learned from Industrial Training Experience

One of the key lessons learned was the importance of understanding the right tool for the right task. WordPress was suitable for fast website and landing page development, while Laravel was more appropriate for building structured systems like DomTrack. This distinction helped improve decision-making in system development and project planning.

The training experience also emphasized the importance of continuous learning. SEO, system development, and real-world deployment tasks required self-learning, experimentation, and guidance from supervisors. Proper testing, documentation, and communication were identified as critical factors in ensuring successful task execution.

CHAPTER 5

CONCLUSION, BENEFITS & RECOMMENDATIONS

5.1 Conclusion

The industrial training at Pemaju Digital Sdn. Bhd. provided extensive exposure to practical IT tasks and real-world software development processes. During the internship, daily responsibilities such as creating WordPress landing pages, learning and applying SEO strategies, and developing automotive prospect systems for brands like Honda, Perodua, Proton, and Toyota using Laragon allowed the application of theoretical knowledge in a professional setting. These tasks improved familiarity with client requirements, website management, and digital marketing strategies.

In addition to daily tasks, involvement in the DomTrack system development strengthened technical skills in Laravel, database management using MySQL and PHPMyAdmin, and version control with Git. Working on DomTrack also emphasized the importance of data integrity, security, and structured workflows, which are critical in a professional environment. The combination of routine tasks and project work provided a well-rounded understanding of IT operations within a company.

The experience also fostered personal and professional growth. Collaborating with team members taught effective communication, teamwork, and time management, while problem-solving challenges helped develop critical thinking and adaptability. Contributing to the DomTrack project and the company's daily operations gave a sense of accomplishment and confidence in handling complex tasks. This internship prepared the intern to transition smoothly into future professional roles in IT, bridging academic knowledge with industry practice.

5.2 Benefits of Industrial Training

The internship delivered significant benefits, both technically and professionally. On the technical side, the intern enhanced proficiency in Laravel framework, database design, and

MySQL management, as well as PHPMyAdmin operations. Exposure to Git strengthened collaborative coding practices, while UI/UX considerations in DomTrack improved the ability to create intuitive interfaces. Learning SEO techniques and managing WordPress websites broadened understanding of digital marketing and client-facing web solutions.

On a professional level, the internship developed essential soft skills such as time management, teamwork, and communication. Observing and participating in company workflows provided insight into project planning, coordination, and execution. The experience encouraged adaptability when learning new tools, reinforced accountability when handling sensitive data, and highlighted the importance of structured documentation and reporting. Collectively, these benefits enhanced the intern's readiness for future professional responsibilities in IT and digital development.

5.3 Recommendations for Future Improvement

For the company, it is recommended to implement further automation for routine tasks, such as report generation or client notifications, to increase efficiency and reduce manual effort. Structured training sessions for interns on tools like Laravel, WordPress, and SEO can improve skill acquisition and accelerate contributions to projects. Maintaining thorough documentation for internal systems and processes will facilitate knowledge transfer and support future maintenance. Additionally, strengthening data security and backup measures will safeguard sensitive client and domain information.

For future interns, active involvement in both technical projects and daily operational tasks is crucial for maximizing learning outcomes. Developing strong communication skills, collaborating effectively with team members, and taking initiative to explore new tools and technologies are essential for professional growth. Keeping a detailed log of daily tasks, challenges, and solutions can help track progress, reflect on learning, and provide material for reporting. Following these recommendations will enhance the internship experience while contributing positively to the company's operational efficiency and long-term goals.

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